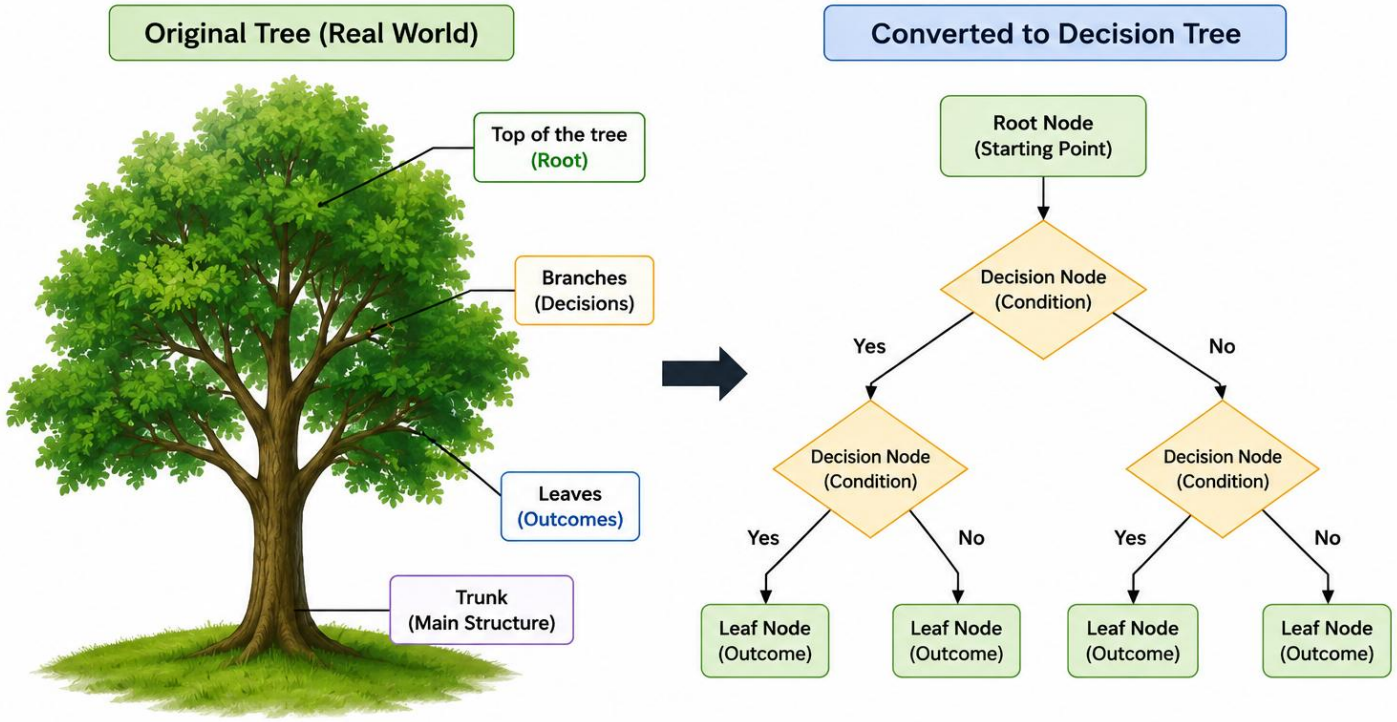
MODULE 3:
Decision tree learning:



What is a Decision Tree?

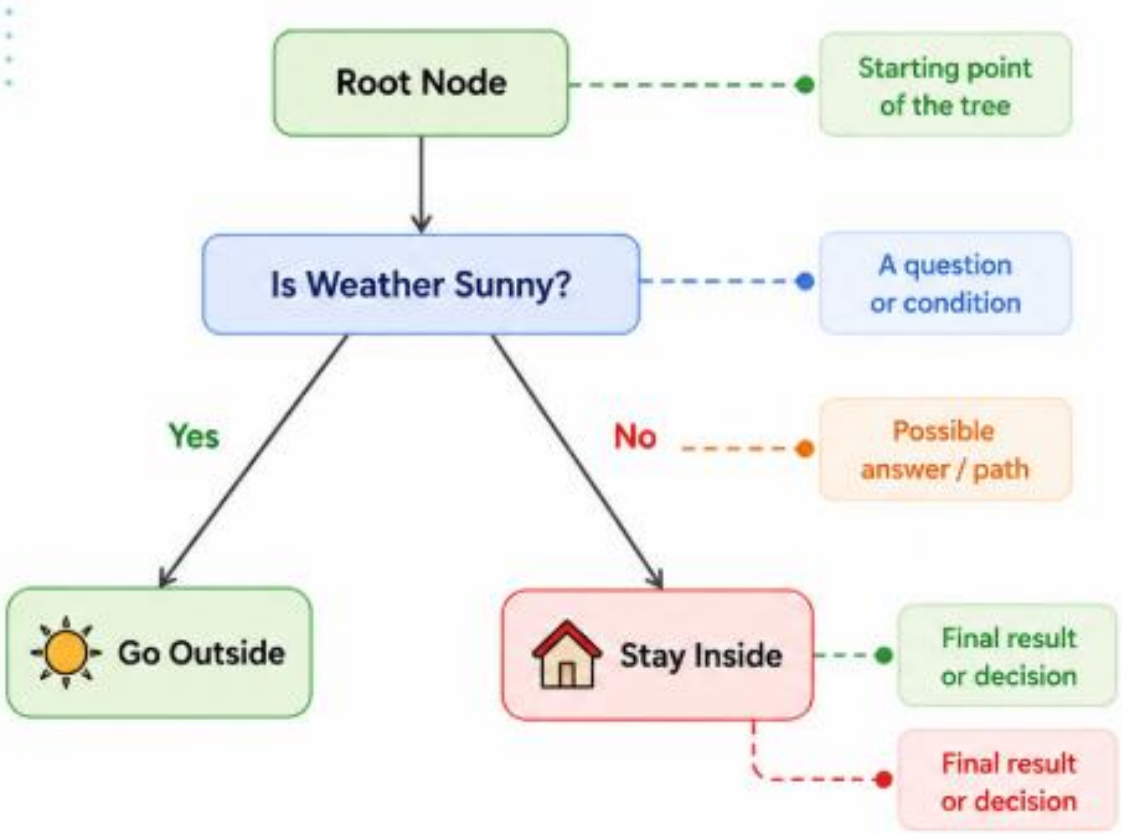
- A Decision Tree is a graphical method used for **decision-making** and **classification**.
- It looks like an **upside-down tree** where
  - each branch represents a **choice** or **condition**
  - each leaf represents a **final outcome.**



Simple Definition:

A Decision Tree is a flowchart-like structure that helps us make decisions by following a series of questions or conditions.

Structure of a Decision Tree



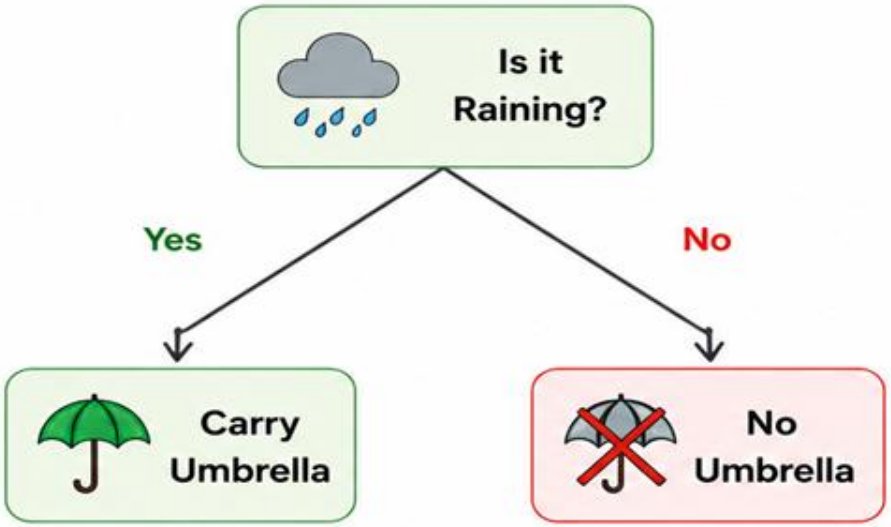
A decision tree starts from a **ROOT** node, ask questions at **DECISION NODES** , follow **BRANCHES** based on answers, and ends at **LEAF** nodes with the final decision.

Components

Component	Meaning
Root Node	Starting point of the tree
Decision Node	A question or condition
Branch	Possible answer/path
Leaf Node	Final result or decision

Real-Life Examples

1. Should I Carry an Umbrella?

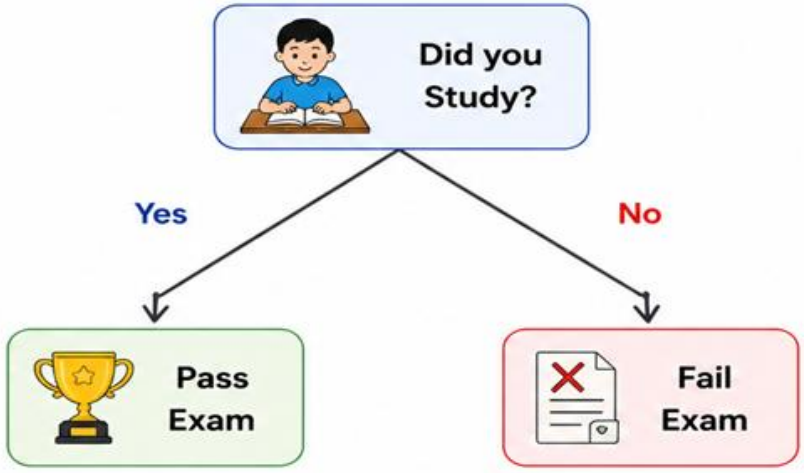




Explanation:

- If it is raining → **Carry Umbrella**
- If it is not raining → **No Umbrella** needed

2. Decision Tree for Exam Result



Interpretation:

- Students who **study** are more likely to **pass**.
- Students who **do not study** are more likely to **fail**.



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Advantages of Decision Trees	Disadvantages of Decision Trees	Applications of De Trees
• Easy to understand	• Can become very large	• Medical diagnosis
• Simple visualization	• May overfit the data	• Weather forecasti
• Supports decision-making	• Sensitive to small changes in data	• Loan approval sys
• Useful for classification and prediction		• Customer classific
• Requires little data preparation		• Student performa prediction
		• Business decision making

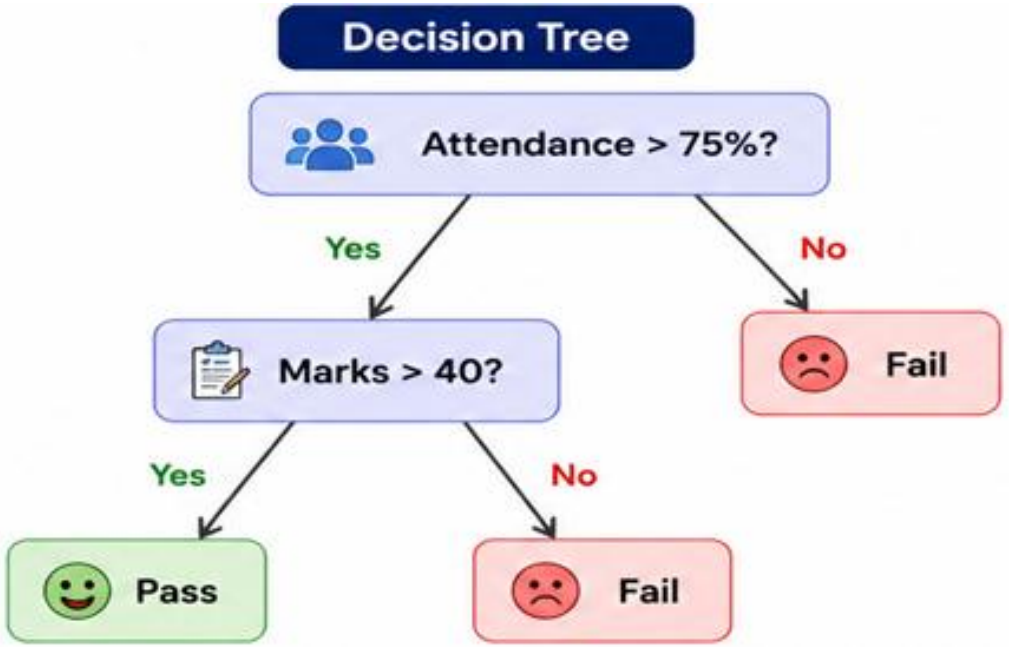
Decision Tree Example in Machine Learning:

Predicting Pass/Fail

Sample Cases

Attendance (%)	Marks	Prediction	Path Followed
80	50	Pass	>75% → >40 → Pass
90	35	Fail	>75% → ≤40 → Fail
60	70	Fail	≤75% → Fail

Decision Trees break down a problem into simple questions. Based on the answers, they predict the final outcome.



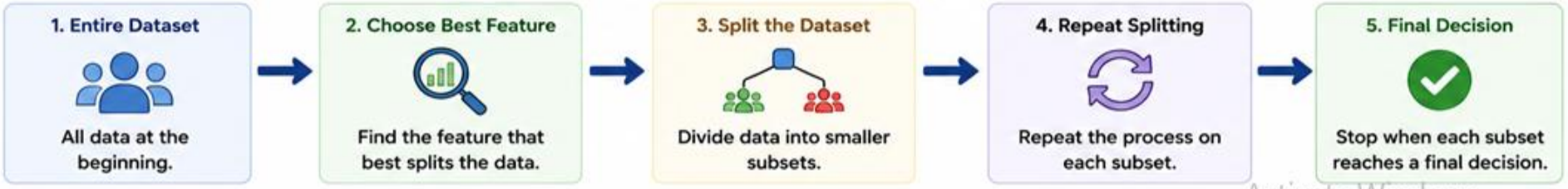
Key Points to Remember

- Decision Tree is a **tree-like model**.
- Starts from a **Root Node**.
- Each **branch represents a decision**.
- Ends at **Leaf Nodes** (final outcomes).
- Used for **classification and prediction**

Divide and Conquer Approach

Decision Trees use the **Divide and Conquer** strategy.

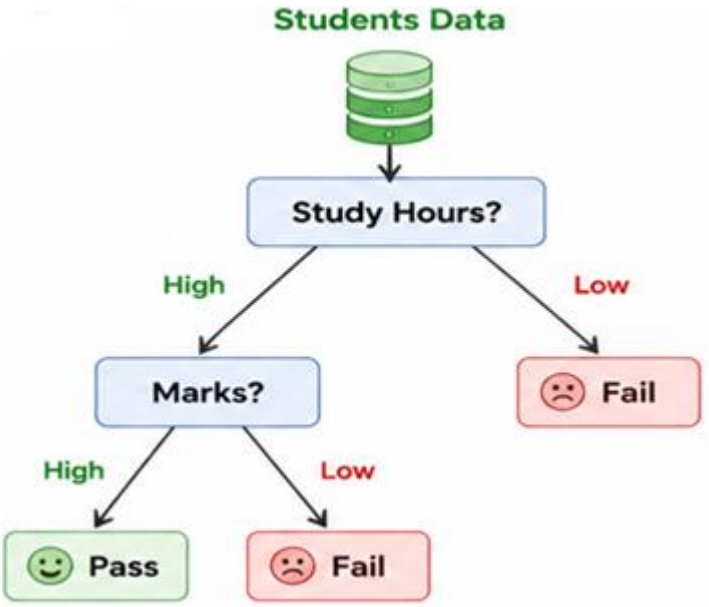
Steps:



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Why Divide and Conquer?

- Breaks large problems into smaller problems.
- Easier to solve.
- Produces accurate decisions.



Advantages	Disadvantages	Applications
• Simplifies complex problems	• Many recursive splits	• Decision Trees
• Faster processing	• Depends on good splitting	• Merge Sort
• Accurate decisions	• Can overfit	• Quick Sort
• Easy to understand	• Higher memory usage	• Binary Search
• Recursive solution	• Increased computation for deep trees	• Image Processing
• Reduces complexity	• Complex implementation for some problems	• Large Data Processing

Attribute Selection in Decision Trees

- **Attribute Selection** is the process of choosing the **best attribute (feature)** to split the dataset at each node of a decision tree.

- The selected attribute should create the most informative split, resulting in smaller and purer subsets of data.

Why is Attribute Selection Needed?

- Different attributes have **different levels of importance**.
- The best attribute creates the **purest child nodes**.
- Proper attribute selection improves **prediction accuracy**.
- It reduces the **number of tree levels**, making the tree simpler.
- It helps build a **more efficient and understandable** decision tree.

Example

- Consider the following dataset to predict **Play Tennis**.

Outlook	Temperature	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No

Which attribute should become the Root Node?

Possible attributes:

- Outlook
- Temperature
- Humidity
- Wind



How is the Best Attribute Selected?

- A mathematical measure is used to identify the best attribute for splitting the dataset.
- Common measures include:
  - **Information Gain** (used in ID3)
  - **Gain Ratio** (used in C4.5/C5.0)
  - **Gini Index** (used in CART)

Key Point

- The attribute with the **highest Information Gain (or Gain Ratio/Gini Index, depending on the algorithm)** is selected as the **Root Node**, because it produces the best split of the data.

Entropy

Entropy - Measuring Uncertainty or Impurity

- Entropy measures the amount of **uncertainty** or **disorder** in a dataset.
  - **High Entropy** → More uncertainty (mixed classes)
  - **Low Entropy** → Less uncertainty (pure classes)

Entropy Formula

- For a dataset **S** with classes C1,C2,…,Cn

$$\text{Entropy}(S) = -\sum_{i=1}^{n} p_i \log_2(p_i)$$

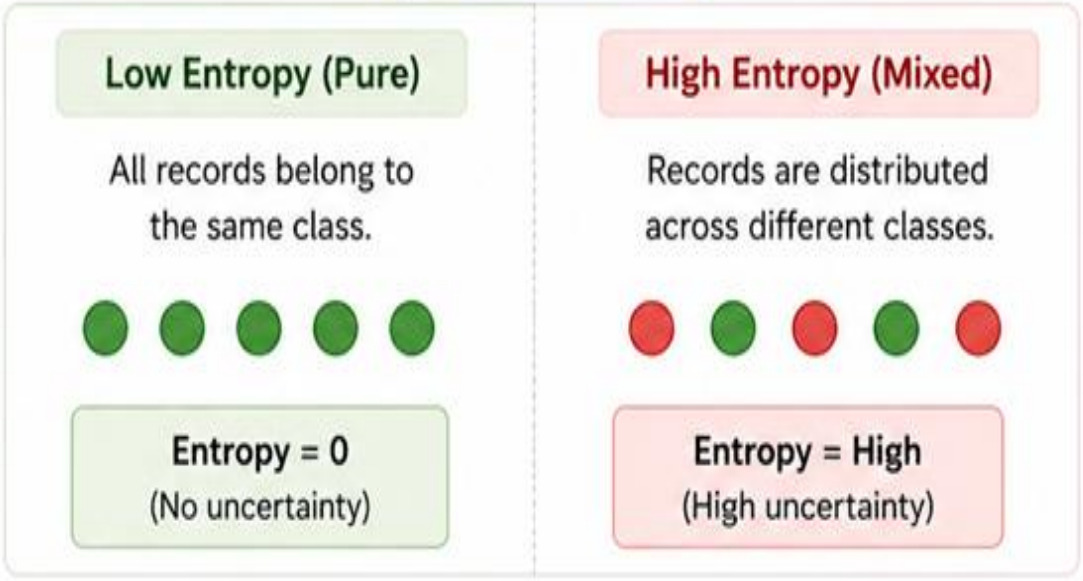
Where:

- pi = Probability of class Ci in dataset S
- log2 = Logarithm to base 2

Note:

- Entropy values range from **0 to log2(n)**
- For **binary classification**, the entropy range is **0 to 1**.

Visual Intuition



"Lower entropy means the dataset is more "pure" and easier to classify."

Dataset Composition	p(Yes)	p(No)	Entropy	Interpretation
10 Yes, 0 No	1.0	0.0	0	Pure (No uncertainty)
5 Yes, 5 No	0.5	0.5	1	Maximum uncertainty
8 Yes, 2 No	0.8	0.2	0.722	Moderate uncertainty

Q1. Consider a dataset containing 10 students: Calculate the entropy.

Result	Number
Pass	8
Fail	2
Total	10

Step 1: Calculate probabilities

$$P(\text{Pass})=8/10=0.8$$

$$P(\text{Fail})=2/10=0.2$$

Step 2: Apply entropy formula

$$\text{Entropy}(S)=-[0.8*\log_2(0.8)+0.2*\log_2(0.2)]$$

$$\text{Entropy}(S)\approx 0.722$$

Interpretation

Entropy = 0.722

So the dataset has **some uncertainty** because it contains both Pass and Fail students.

Q2. Consider a dataset Calculate the entropy.

Pass	Fail	Entropy
10	0	?
5	5	?
9	1	?

- Answers:
- **10, 0 → 0**
- **5, 5 → 1**
- **9, 1 → 0.469**
- **Pure dataset → Entropy = 0**
- **Completely mixed binary dataset → Entropy = 1**

Q3. Calculate Information Gain for ONE Attribute

Predict whether a student will Pass based on Attendance.

Student	Attendance	Result
1	High	Pass
2	High	Pass
3	High	Pass
4	High	Fail
5	Low	Fail
6	Low	Fail

So:

- Total = 6
- Pass = 3
- Fail = 3

Step 1: Entropy of the complete dataset

$$\text{Entropy}(S) = -(3/6)\log_2(3/6) - (3/6)\log_2(3/6)$$

$$\text{Entropy}(S) = 1$$

Step 2: Split according to Attendance

a)High

High attendance:

- Pass = 3
- Fail = 1

$$\text{Entropy}(\text{High}) = -(3/4)\log_2(3/4) - (1/4)\log_2(1/4)$$

$$= 0.811$$

b)Low

Low attendance:

- Pass = 0
- Fail = 2

$$\text{Entropy}(\text{Low}) = 0$$

because the subset is completely pure.

Step 3: Weighted Entropy

- Weighted Entropy=(4/6)*(0.811)+(2/6)*(0)

  =0.541

Step 4: Information Gain

- IG=Entropy(S)−Weighted Entropy
- IG=1−0.541
- IG=0.459

Information Gain = reduction in entropy after splitting.

Q4. Predict whether a student will pass.

Student	Attendance	Assignment	Result
1	High	Yes	Pass
2	High	Yes	Pass
3	High	No	Pass
4	High	No	Fail
5	Low	Yes	Fail
6	Low	No	Fail

Which is better?

/ \

Attendance Assignment

| |

IG=? IG=?

Assignment has:

- Yes
- No

<u>Assignment = Yes</u>

Attendance	Assignment	Result
High	Yes	Pass
High	Yes	Pass
Low	Yes	Fail

Therefore:

- Pass = 2
- Fail = 1

Entropy(Yes)=$-[2/3*\log_2(2/3)+1/3*\log_2(1/3)]$

Entropy(Yes)= 0.918

Assignment = No

Attendance	Assignment	Result
High	No	Pass
High	No	Fail
Low	No	Fail

Therefore:

- Pass = 1
- Fail = 2

Entropy(NO)= 0.918

Weighted Entropy for Assignment

- There are 3 records for Yes and 3 records for No.

Weighted Entropy(Assignment)= 3/6*(0.918)+3/6*(0.918)

=0.918

Information Gain for Assignment:

IG(Assignment)=Entropy(S)−Weighted Entropy

=1−0.918

IG(Assignment)=0.082

Final Comparison

- Now give students the important comparison.

Feature	Weighted Entropy	Information Gain
Attendance	**0.541**	**0.459**
Assignment	0.918	0.082

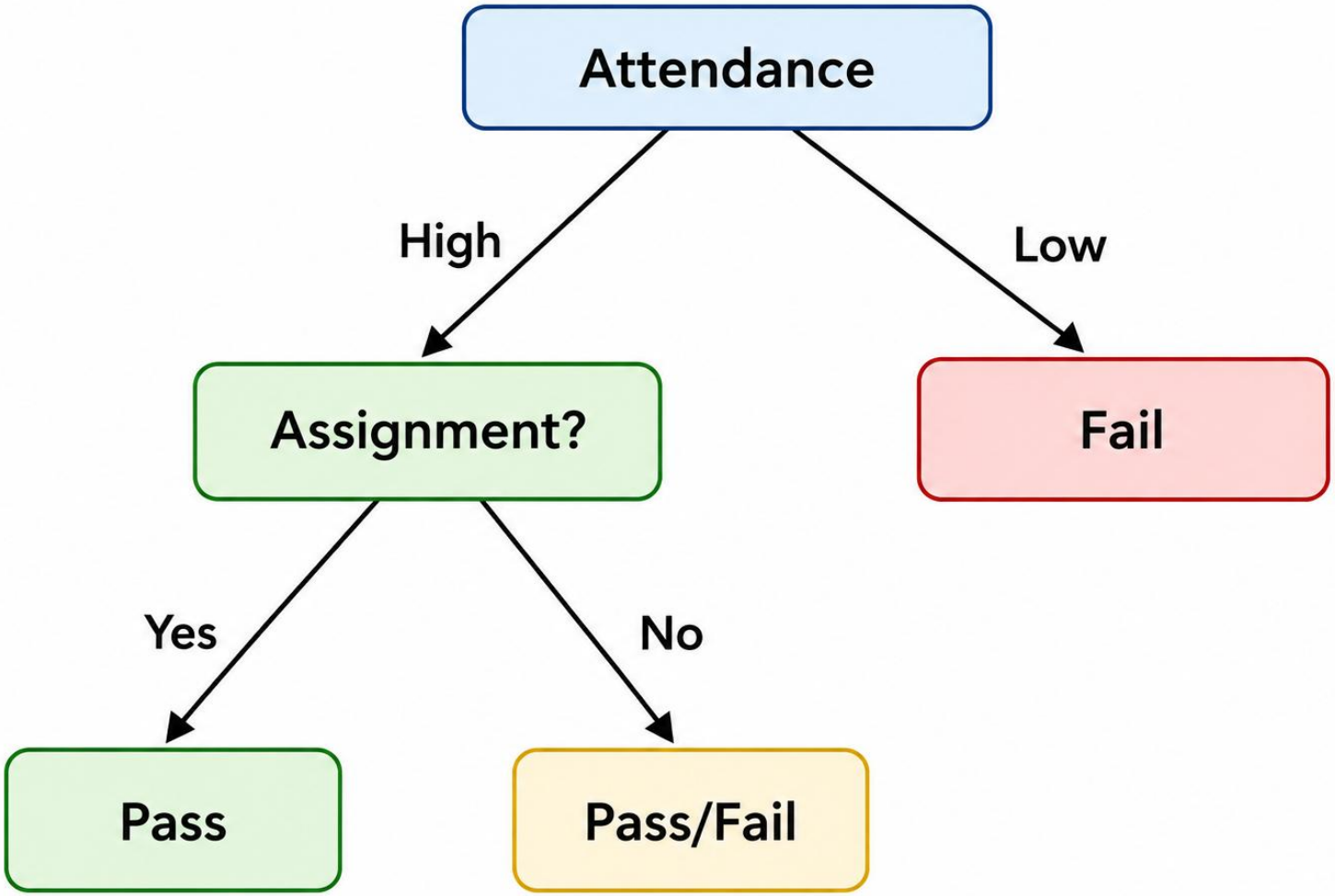
- **Which feature should become the root?**

$$0.459 > 0.082$$

Therefore:

- **Attendance is selected as the root node.**

Resulting Decision Tree:



Information Gain:

- **Information Gain (IG)** measures how much the uncertainty/impurity of a dataset is reduced when it is split using a particular attribute.
- **Information Gain tells us how useful an attribute is for splitting the data.**
- The attribute with the **highest Information Gain** is selected as the splitting attribute.
- **Formula:**

$$IG(S,A)=Entropy(S)-Weighted\ Entropy(S,A)$$

where:

- S= original dataset
- A= attribute used for splitting

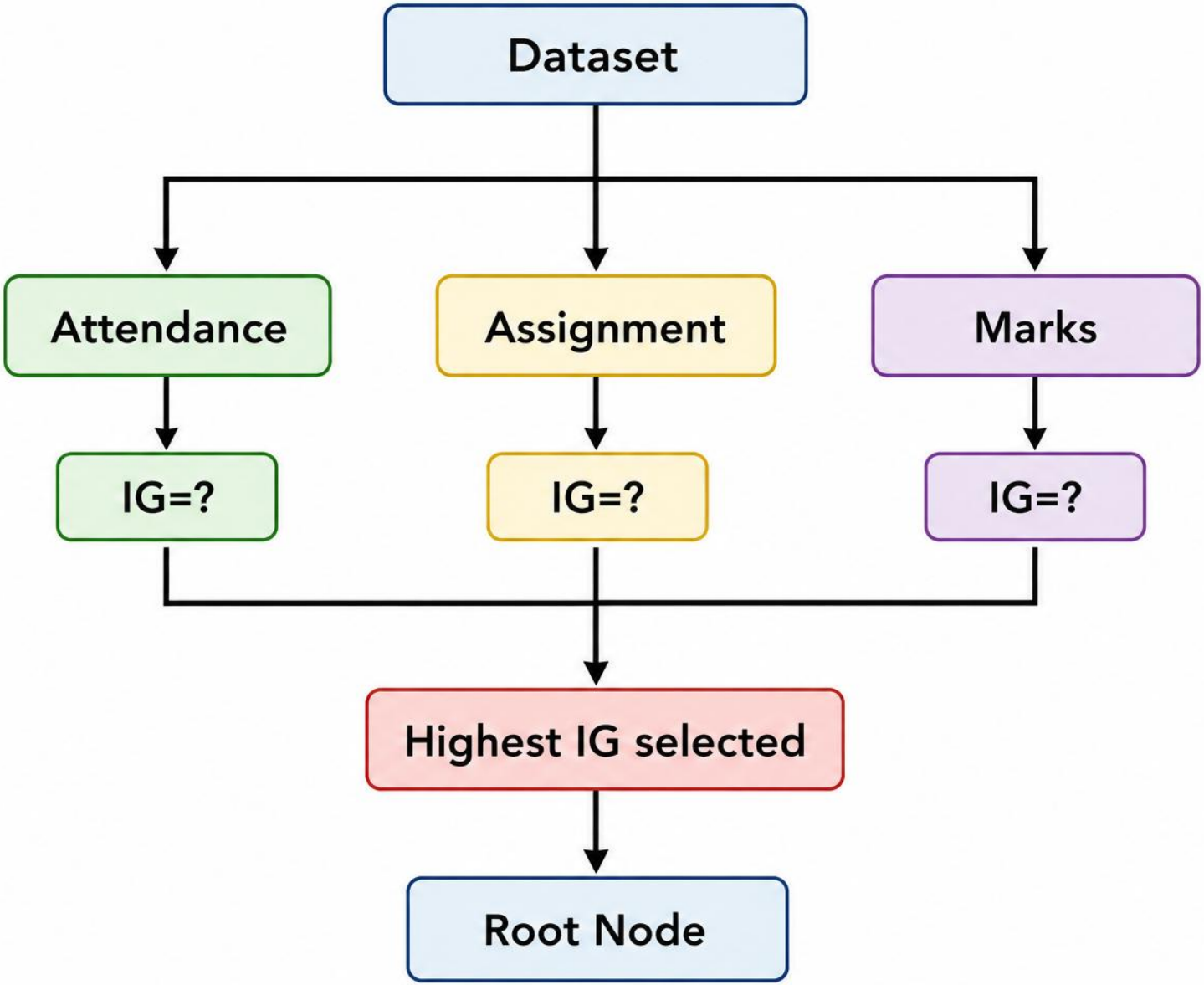
Why Information Gain is Needed?

Suppose we have several attributes:

- Attendance
- Assignment
- Internal Marks
- Study Hours

We need to decide:

- **Which attribute should become the root node?**
- Entropy alone cannot answer this.
- Information Gain helps us compare the attributes.


Dataset
Attendance
Assignment
Marks
IG=?
IG=?
IG=?
Highest IG selected
Root Node

Attribute Selection

When multiple attributes are available:

Procedure:

- Calculate the entropy of the complete dataset.
- Select an attribute.
- Divide the dataset according to its values.
- Calculate entropy for each subset.
- Calculate weighted entropy.
- Calculate Information Gain.
- Repeat for all candidate attributes.
- Select the attribute with the **highest Information Gain**.

Q5. Play Tennis Dataset

Outlook	Temperature	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No

Which attribute should become the root?

Possible attributes:

- Outlook
- Temperature
- Humidity
- Wind

Step 1 – Calculate Initial Entropy

- There are:
  - Yes = 3
  - No = 3
- Therefore: Entropy(S)=1

Step 2 – Calculate Information Gain

Attribute	Weighted Entropy	Information Gain
Outlook	0.459	**0.541**
Temperature	0.792	0.208
Humidity	1.000	0.000
Wind	0.541	0.459

Therefore:

Outlook has the highest Information Gain

So: **Outlook becomes the Root Node.**

Build the Tree Step-by-Step:

- **Outlook=Root**

Outlook

**/ | **

Sunny Overcast Rain

Sunny

Records:

Outlook	Play
Sunny	No
Sunny	No

- **Entropy = 0**

Therefore:

- **Sunny → No**
- **It becomes a leaf.**

Overcast

Records:

Outlook	Play
Overcast	Yes

- **Entropy = 0**

Therefore:

- **Overcast → Yes**
- **It becomes a leaf.**

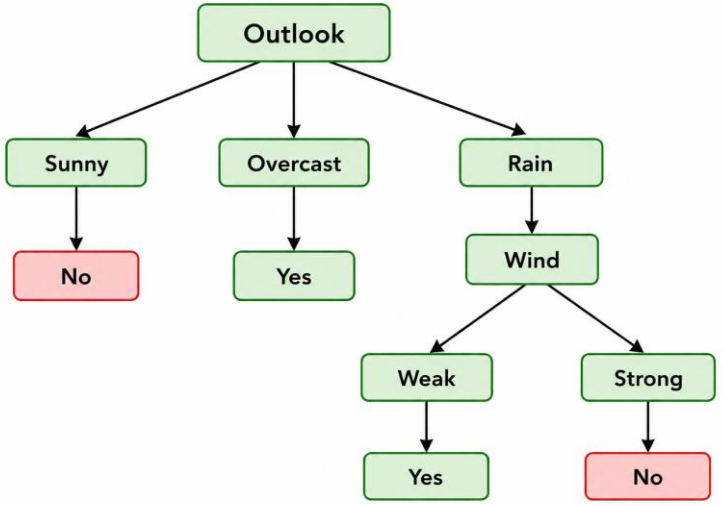
Rain

Records:

Outlook	Temperature	Humidity	Wind	Play
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No

This subset is not pure.

Which attribute should we use to split the Rain subset?



Gain Ratio

- **Gain Ratio** is an improved measure used to reduce the bias of Information Gain toward attributes with many distinct values.

$$Gain\ Ratio = \frac{Information\ Gain}{Split\ Information}$$

Split Information

$$SplitInfo(S, A) = -\sum_{v} \frac{|S_v|}{|S|} \log_2 \frac{|S_v|}{|S|}$$

- Higher Gain Ratio → Better attribute

Information Gain vs Gain Ratio

Information Gain

- Measures reduction in entropy
- Can favour many-valued attributes
- Used in ID3
- Higher is better

Gain Ratio

- Normalizes Information Gain
- Reduces this bias
- Used in C4.5 and related tree-learning approaches
- Higher is better

C5.0 Decision Tree Algorithm

- **C5.0** is a **supervised machine learning algorithm** used mainly for **classification**.
- It builds a **decision tree** by selecting suitable attributes to divide the training data into meaningful groups.
- **C5.0 is a decision tree learning algorithm that builds a classification tree by selecting appropriate attributes, handling missing values, and pruning unnecessary branches.**
- It is an improved successor to **C4.5**.

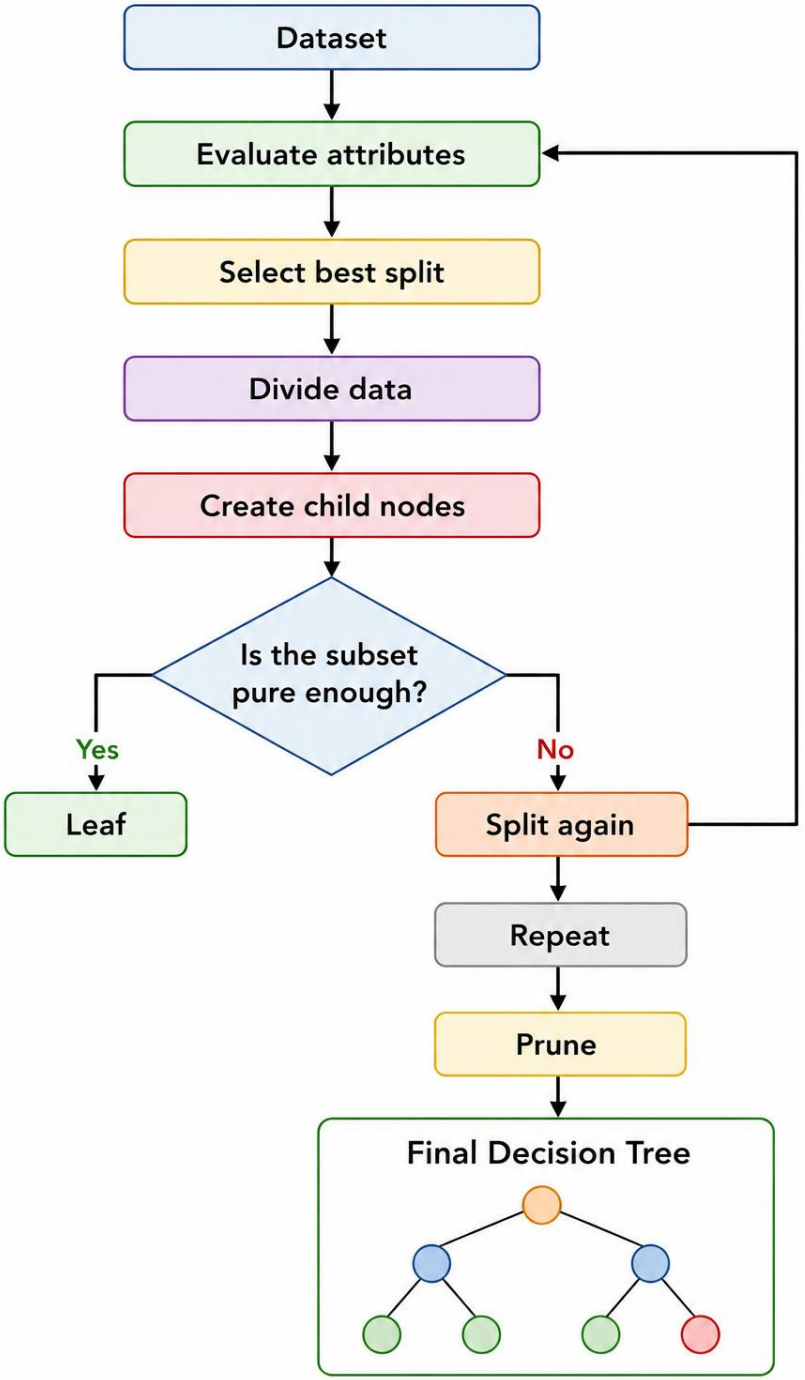
Why C5.0?

Algorithm	Main Feature
ID3	Information Gain
C4.5	Gain Ratio + pruning
C5.0	Improvements in speed, memory usage, pruning and rule generation

Main Characteristics of C5.0

- Build decision trees for classification.
- Handle **categorical attributes**.
- Handle **continuous/numerical attributes**
- Handle **missing values**.
- Perform **tree pruning**.
- Generate **classification rules**.
- Produce relatively compact trees.
- Work efficiently with large datasets.

Basic Idea of C5.0



Advantages of C5.0

- Easy-to-understand decision tree representation.
- Supports classification.
- Handles categorical and numerical data.
- Handles missing values.
- Performs pruning.
- Can generate classification rules.
- Efficient for large datasets.
- Usually produces relatively compact trees.

Limitations of C5.0

- Can still overfit if not appropriately controlled.
- Tree structure can change with changes in training data.
- Complex datasets may produce complicated trees.
- Attribute selection can affect the final tree.
- Interpretation becomes harder when the tree becomes very large.

C5.0 vs ID3 vs C4.5

Feature	ID3	C4.5	C5.0
Classification	✓	✓	✓
Information Gain	✓	✓	✓/related splitting criteria
Gain Ratio	✗	✓	Improved splitting approach
Continuous attributes	Limited	✓	✓
Missing values	Limited	✓	✓
Pruning	Limited	✓	✓
Rule generation	Limited	✓	✓
Efficiency	Lower	Moderate	Generally higher

PRUNING THE DECISION TREE

- Pruning is the process of **removing branches or nodes** from a decision tree that provide little or no improvement in prediction accuracy.
- Instead of allowing the tree to grow indefinitely, pruning **simplifies the structure** by eliminating branches that represent noise or unimportant patterns.
- **Pruning is the technique of reducing the size of a decision tree by removing unnecessary branches to improve prediction accuracy and reduce overfitting.**

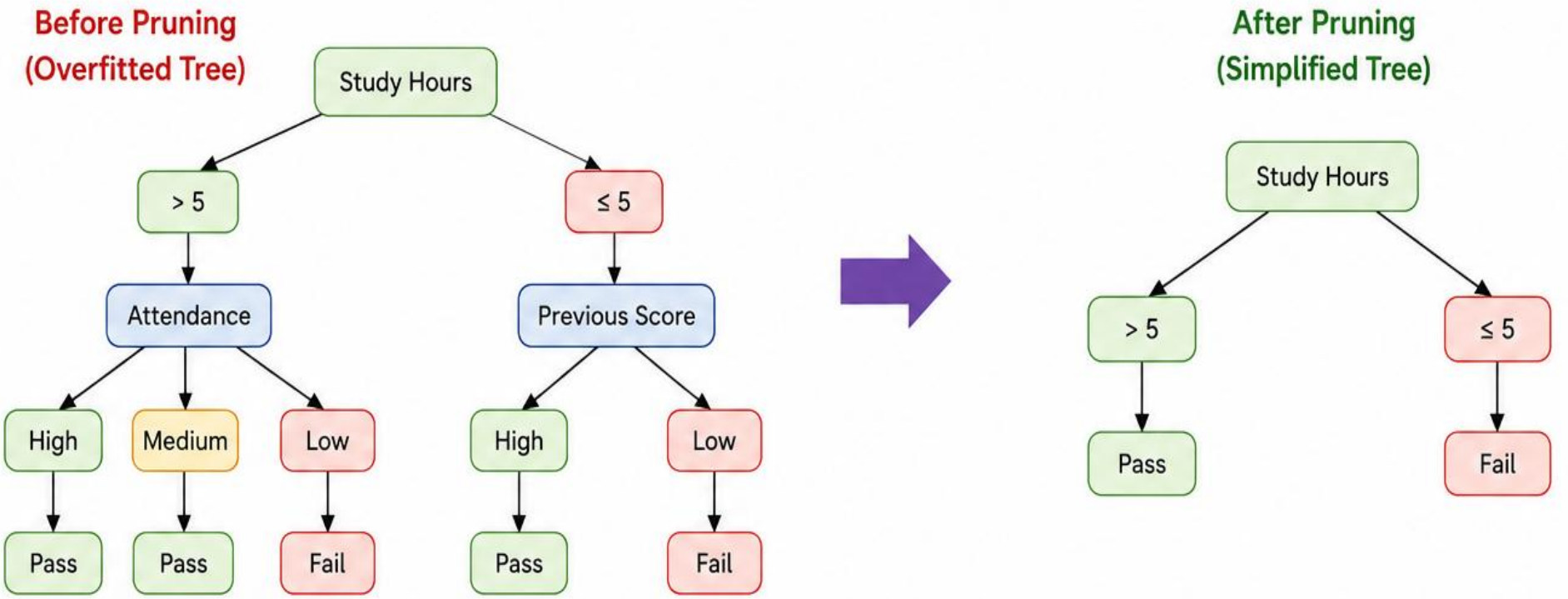
WHY PRUNING IS NEEDED?

- Reduces Overfitting
- Improves Accuracy
- Produces Smaller Tree
- Easy to Understand
- Reduces Complexity
- Better Generalization

BENEFITS OF PRUNING

- Prevents overfitting
- Simplifies the decision tree
- Removes noisy or unimportant branches
- Improves prediction on unseen data
- Faster classification
- Lower storage requirements

Example of Pruning



Types of Pruning:

- Pre-Pruning (Early Stopping)
- Post-Pruning (Late Pruning)

1. Pre-Pruning (Early Stopping)

- Pre-pruning stops the tree from growing before it becomes too large.

- Instead of splitting every node, the algorithm checks whether the split is worthwhile. If not, it stops splitting.



Common Stopping Conditions

- Maximum tree depth reached
- Minimum number of samples in a node
- Minimum information gain
- Minimum reduction in impurity
- Maximum number of leaf nodes

Advantages

- Faster tree construction
- Prevents unnecessary branches
- Lower memory usage
- Simple implementation

Disadvantages

- Important branches may be removed too early.
- Can produce an under-fitted model.

Post-Pruning (Late Pruning)

- In post-pruning, the tree is allowed to **grow completely first.**
- After the full tree is built, **unnecessary branches are removed.**
- This is the **most commonly used pruning method.**



Steps:

Step 1: Build the complete decision tree.

Step 2: Evaluate each branch.

Step 3: Remove branches that do not improve accuracy.

Step 4: Keep the simplified tree.

Advantages

- Produces better generalization
- Higher prediction accuracy
- Reduces overfitting effectively

Disadvantages

- Takes more computation time
- Requires building the full tree first

COMPARISON OF PRE-PRUNING AND POST-PRUNING

Feature	Pre-Pruning	Post-Pruning
When applied	Before tree is fully grown	After tree is completely grown
Speed	Faster	Slower
Complexity	Low	Higher
Risk	Underfitting	Less risk of underfitting
Accuracy	Moderate	Usually higher
Memory usage	Lower	Higher during training

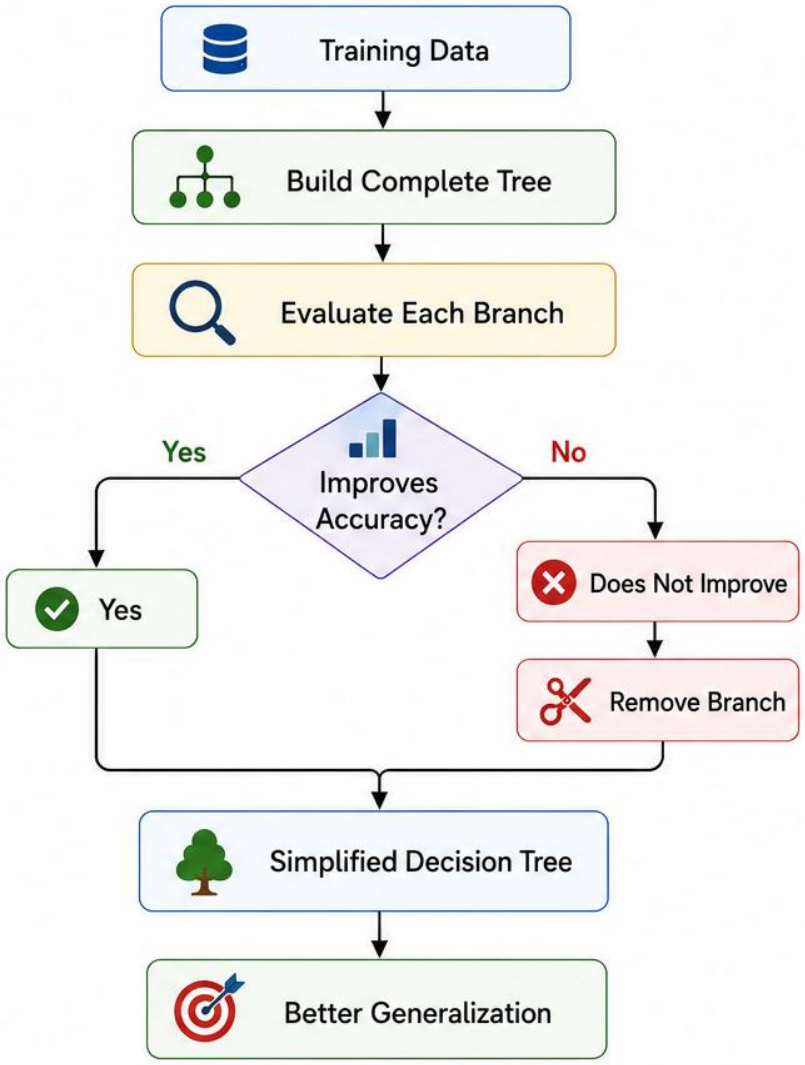
Limitations of Pruning

- Excessive pruning may reduce accuracy.
- Selecting the best pruning level is difficult.
- Pre-pruning may stop useful splits.
- Post-pruning requires additional computation.

Applications

- Decision tree pruning is widely used in:
- Medical diagnosis
- Credit risk assessment
- Loan approval systems
- Fraud detection
- Customer churn prediction
- Student performance analysis
- Sales forecasting
- Marketing analytics

FLOWCHART OF DECISION TREE PRUNING



KEY POINTS

1. Pruning removes unnecessary branches from a decision tree.
2. Its main purpose is to **reduce overfitting**.
3. **Two types of pruning:**
   - **Pre-Pruning** (stops tree growth early)
   - **Post-Pruning** (removes branches after the tree is built)
4. C5.0 uses **automatic post-pruning** to improve prediction accuracy.
5. Pruning creates **smaller, simpler**, and **more reliable** decision trees.

Classification Rules Learning

- A **classification rule** is a rule in the form:

IF condition THEN class

- The **IF part** contains one or more conditions, while the **THEN part** specifies the class or outcome.
- **General Form**
  - IF <condition>
  - THEN <class>

Example

- Suppose we want to predict whether a student will **Pass** or **Fail**.

IF Attendance = High

AND Internal Marks = High

THEN Result = Pass

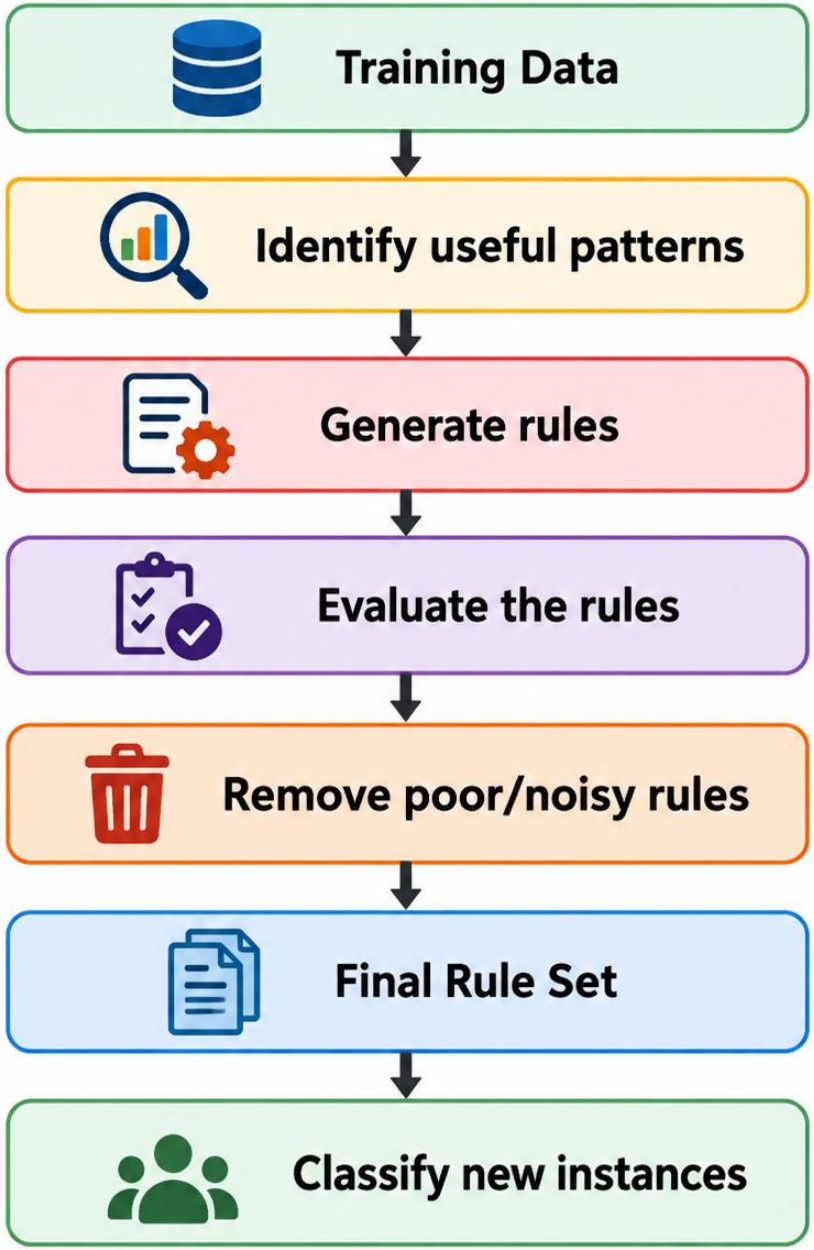
Components of a Classification Rule:

- **Antecedent:** Conditions in the IF part.
- **Consequent:** Class in the THEN part.

Important Characteristics

- Rules are easy to understand.
- Each rule represents a pattern in the data.
- Multiple rules can be used to classify different instances.
- Rules can be generated directly from training data or derived from a decision tree.

Classification Rule Learning Process:



Example

- Consider:

Attendance	Internal Marks	Result
High	High	Pass
High	Low	Pass
Low	High	Fail
Low	Low	Fail

- The system can learn:

IF Attendance = High

THEN Result = Pass

IF Attendance = Low

THEN Result = Fail

Separate and Conquer Approach:

- The **Separate and Conquer approach** is a method for learning classification rules.
- It learns **one rule at a time**. After a rule is created, the training examples covered by that rule are removed. The process continues until the required examples have been classified.
- It is also called the **Covering approach**.
- **Learn a rule → Remove the examples covered by the rule → Learn the next rule.**

Steps of Separate and Conquer:

Step 1: Start with the complete training dataset

Training Data

↓

All examples

Step 2: Learn a rule

- Find conditions that correctly identify a particular class.

IF Attendance = High

THEN Result = Pass

Step 3: Remove covered examples

- Remove the examples correctly covered by the rule.

Training Data

↓

Learn Rule

↓

Remove Covered Examples

Step 4: Learn another rule

- Use the remaining examples to learn the next rule.

IF Attendance = Low

THEN Result = Fail

Step 5: Repeat

- Continue learning rules until the remaining examples are classified or no useful rule can be generated.

Example:

- Suppose we have:

Student	Attendance	Marks	Result
A	High	High	Pass
B	High	Low	Pass
C	Low	High	Fail
D	Low	Low	Fail

First Rule

IF Attendance = High
THEN Result = Pass

This rule covers:
A → Pass
B → Pass

These examples are removed.

Remaining Data
C → Fail
D → Fail

Second Rule

IF Attendance = Low
THEN Result = Fail

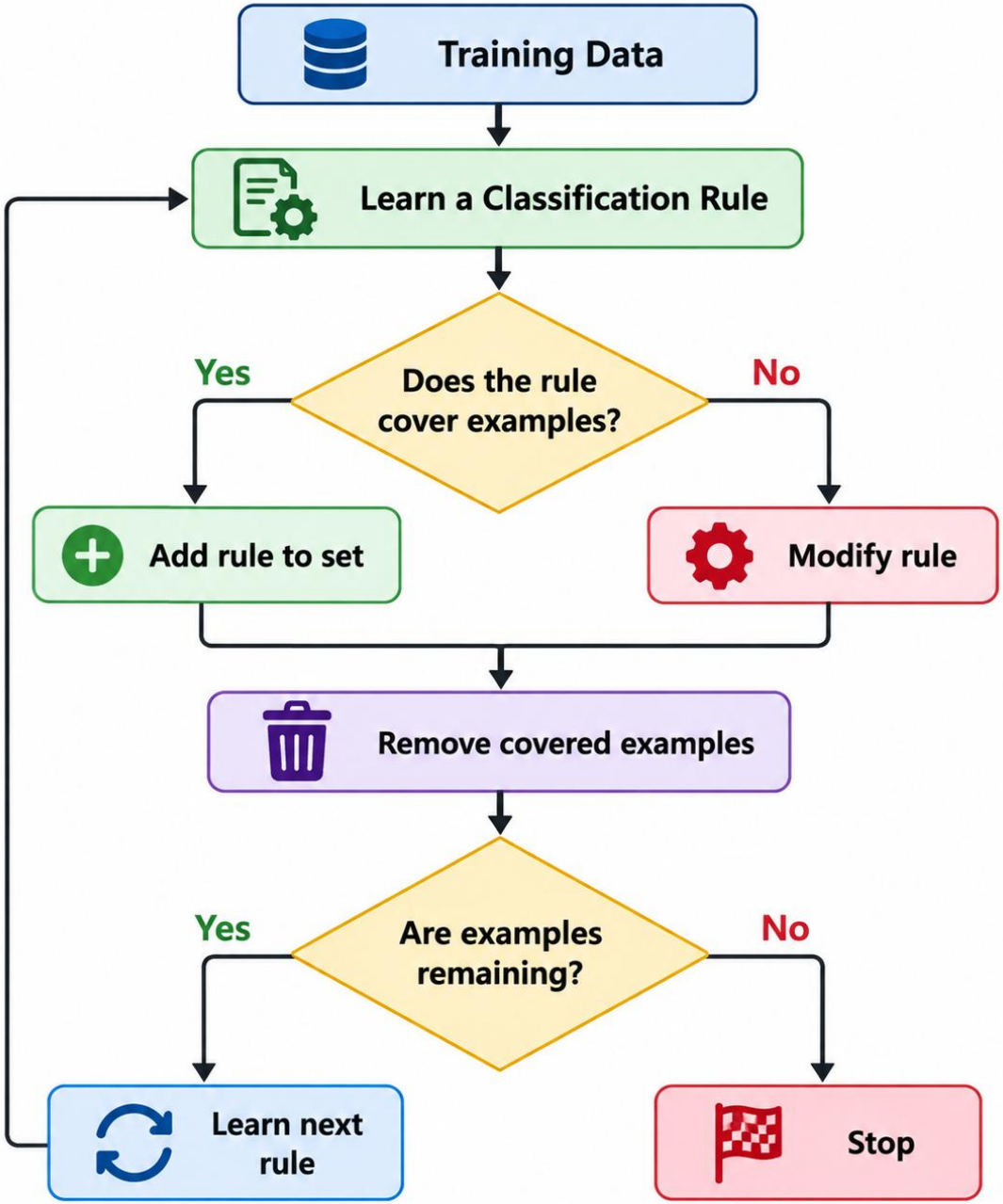
The remaining examples are now covered.

Final Rule Set

Rule 1:
IF Attendance = High
THEN Result = Pass

Rule 2:
IF Attendance = Low
THEN Result = Fail

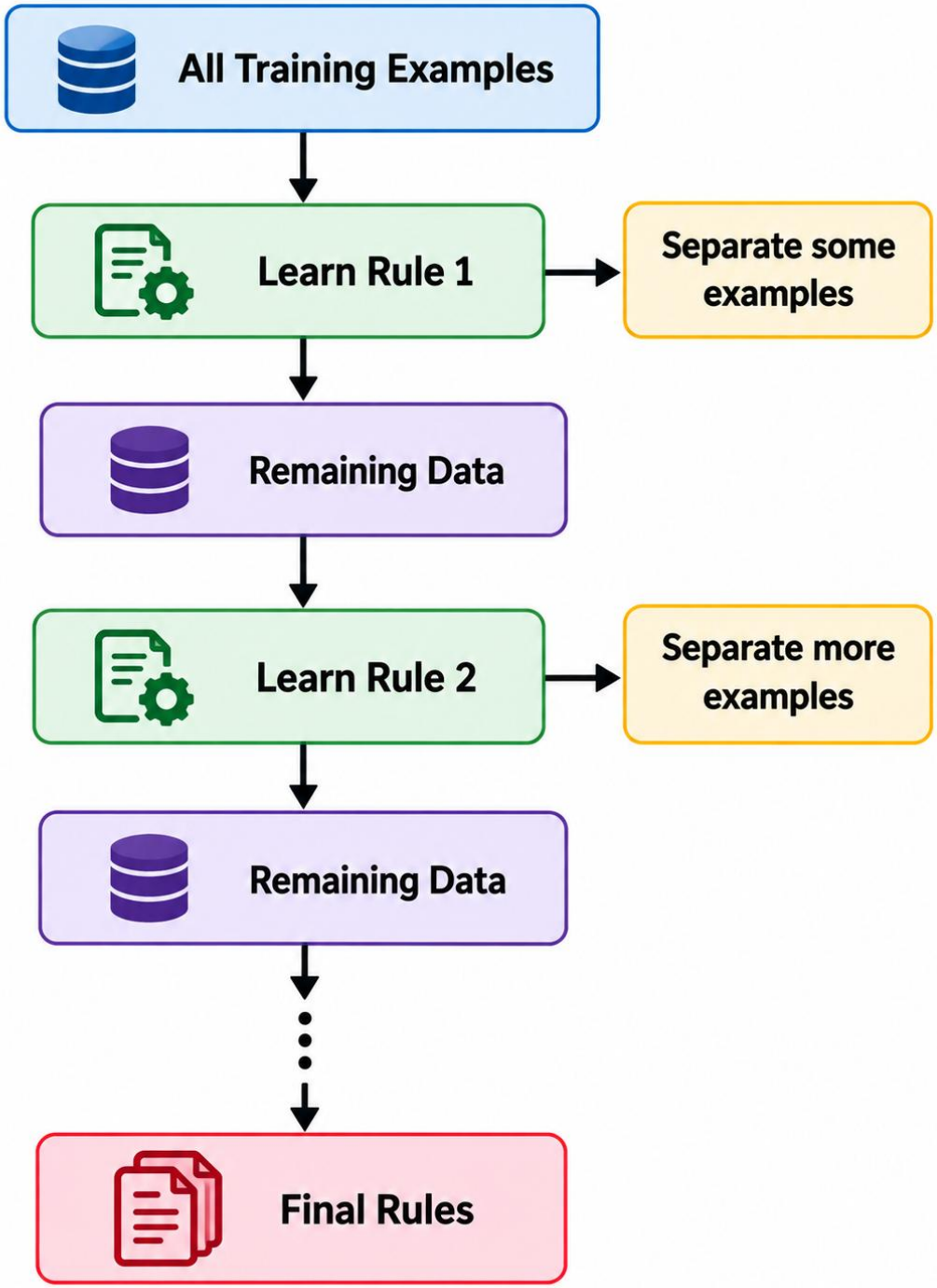
Flowchart:


Training Data
Learn a Classification Rule
Does the rule cover examples?
Yes
No
Add rule to set
Modify rule
Remove covered examples
Are examples remaining?
Yes
No
Learn next rule
Stop

Why is it called "Separate and Conquer"?

The name describes the strategy:

- **Separate:** A rule **separates** examples belonging to one class from the others.
- **Conquer:** Once those examples are handled, they are removed, and the algorithm **conquers the remaining examples** by learning additional rules.


All Training Examples
Learn Rule 1
Separate some examples
Remaining Data
Learn Rule 2
Separate more examples
Remaining Data
Final Rules

Advantages of Separate and Conquer

- Produces **easy-to-understand rules**.
- Rules can be interpreted directly.
- Learns rules **one at a time**.
- Can focus on specific classes.
- Removes already classified examples.
- Useful for classification problems.

Limitation

- A poor early rule may affect the examples available for later rules.
- The order of rules can influence the final rule set.
- It may require several iterations to generate a good set of rules.

The 1R Algorithm

- 1R (One Rule) is a simple **classification algorithm** that generates a **one-level classification rule** using **only one attribute.**
- The algorithm **examines each attribute** and **creates a rule based on that attribute.**
- It then selects the attribute whose rule produces the **fewest classification errors.**
- **1R is a simple classification algorithm that uses a single attribute to construct a one-rule classifier.**
- **One attribute → One set of rules → Classification**

Basic Idea of 1R

- The 1R algorithm assumes that **one attribute is sufficient to classify the data reasonably well**.
- For each attribute:
  - Divide the training data according to the values of that attribute.
  - For each attribute value, find the **most frequent class**.
  - Create a rule assigning that class to the attribute value.
  - Count the classification errors.
  - Repeat for all attributes.
  - Select the attribute with the **minimum number of errors**.

Example

- Consider the following dataset for predicting whether a student will **Pass or Fail**.

Student	Attendance	Assignment	Result
A	High	Yes	Pass
B	High	No	Pass
C	Low	Yes	Fail
D	Low	No	Fail
E	High	Yes	Pass
F	Low	Yes	Fail

- Suppose 1R considers the **Attendance** attribute.
  - **Attendance = High**

Result	Count
Pass	3
Fail	0

Most frequent class = Pass

Rule: IF Attendance = High, THEN Result = Pass

– **Attendance = Low**

Result	Count
Pass	0
Fail	3

Most frequent class = Fail

Rule: IF Attendance = Low, THEN Result = Fail

Therefore, the complete rule is:

IF Attendance = High THEN Result = Pass

IF Attendance = Low THEN Result = Fail

Since these rules correctly classify all six examples, the number of errors is 0.

Steps of the 1R Algorithm

- **Step 1: Consider each attribute**
  - Select one attribute at a time.
    - Attendance
    - Assignment
    - Internal Marks
- **Step 2: Group examples by attribute values**
  - For example:
    - Attendance = High
    - Attendance = Low
- **Step 3: Find the most frequent class**
  - For each attribute value, count how many examples belong to each class.
  - Choose the class occurring most frequently.

- **Step 4: Construct rules**
  - Create an IF–THEN rule for each attribute value.
  - IF Attendance = High THEN Pass
- **Step 5: Calculate errors**
  - Count the number of training examples incorrectly classified by the rules.
- **Step 6: Compare attributes**
  - Repeat the process for every attribute.
- **Step 7: Select the best attribute**
  - Choose the attribute that produces the **lowest number of errors**.
- **Step 8: Use the selected rule for classification**
  - The resulting one-rule classifier is used to classify new instances.

- **Selection of the Best Attribute**
  - The best attribute is the one that produces the **minimum classification error**.

Attribute	Number of Errors
Attendance	0
Assignment	1
Internal Marks	2
Study Hours	3

- **Best Attribute = Attendance**
  - The final classifier uses only **Attendance**.

Advantages of 1R

- Very simple and easy to understand.
- Easy to implement.
- Fast to construct.
- Requires only one attribute.
- Produces easily interpretable rules.
- Useful as a baseline classification method.
- Can perform surprisingly well when one attribute strongly determines the class.

Limitations of 1R

- Uses only one attribute.
- May ignore useful information from other attributes.
- Can perform poorly when classification depends on several attributes.
- Sensitive to noisy data.
- It may be less accurate than more complex classification algorithms.

Applications of 1R

- Student performance prediction
- Customer classification
- Simple medical classification
- Product classification
- Basic credit classification
- Preliminary data analysis

REGRESSION METHODS

- Regression is an important **supervised learning technique** used to predict a **continuous numerical value** from one or more input variables.
- The topics covered are:
  - Concept of Regression
  - Simple Linear Regression
  - Ordinary Least Squares Estimation
  - Correlations
  - Multiple Linear Regression

CONCEPT OF REGRESSION

- **Regression** is a statistical and machine learning technique used to study the relationship between a **dependent variable** and one or more **independent variables**, and to predict the value of the dependent variable.

Example:

- Suppose we want to predict the **price of a house** based on its **area**.
- House Area → Regression Model → House Price
    - Independent variable (X): House area
    - Dependent variable (Y): House price
    - If the area increases, the price may also increase.

Examples of Regression Problems:

Input	Output
House area	House price
Study hours	Exam marks
Years of experience	Salary
Temperature	Electricity consumption
Advertising expenditure	Sales

Regression vs Classification:

Regression	Classification
Predicts continuous values	Predicts categorical classes
Output is numerical	Output is a class/label
Example: ₹50,000 salary	Example: Pass/Fail
Example: House price	Example: Spam/Not Spam

SIMPLE LINEAR REGRESSION:

- **Simple Linear Regression** is a regression method that models the relationship between **one independent variable (X)** and **one dependent variable (Y)** using a straight line.
- **Regression Equation:**

$$\hat{Y} = b_0 + b_1 X$$

- Where:
  - $\hat{y}$= Predicted value of Y
  - X = Independent variable
  - b0 = Intercept
  - b1 = Slope of the regression line

Meaning of Intercept and Slope

1. Intercept (b0)

- The intercept is the predicted value of **Y when X = 0**.

2. Slope (b1)

- The slope represents the **change in Y for a one-unit increase in X**.

Example

Suppose the regression equation is: $\hat{y}$ = 20 + 5X

- If X = 4:

  $\hat{y}$=20+5(4)= 40

- Therefore, the predicted value is **40**.

Example: Study Hours and Marks

- Suppose:
  - X = Study Hours
  - Y = Exam Marks
- A regression model may be: $\hat{y}$= 30 + 8X This means:
  - Intercept = 30
  - Slope = 8
- For every additional study hour, predicted marks increase by approximately **8 marks**.

Q1. Find the Regression Equation

- The following data represent the **hours studied (X)** and **marks obtained (Y)** by five students.

Hours Studied (X)	Marks (Y)
1	35
2	40
3	50
4	55
5	65

Find:

- The regression equation
- The slope b1.
- The intercept b0.
- Predict the marks for a student who studies **6 hours**.

Step 1: Calculate the means

$$\bar{x} = \frac{(1+2+3+4+5)}{5} = 3$$

$$\bar{y} = \frac{(35+40+50+55+65)}{5} = 49$$

Step 2: Calculate slope

- Formula: $b1 = \frac{\sum(X-\bar{x})\,(Y-\bar{y})}{\sum(X-\bar{x})^2}$
- Create the calculation table:

X	Y	$X - \bar{x}$	$Y - \bar{y}$	Product	$(X - \bar{x})^2$
1	35	-2	-14	28	4
2	40	-1	-9	9	1
3	50	0	1	0	0
4	55	1	6	6	1
5	65	2	16	32	4
Total				**75**	**10**

Therefore, $b1=\frac{75}{10}=7.5$

Step 3: Calculate intercept

$b0=\bar{y}-b1\bar{x}$

$b0=49-(7.5)(3)$

$b0=26.5$

Step 4: Regression equation

$\hat{y} = 26.5 + 7.5X$

Step 5: Predict marks for 6 hours

$\hat{y} =26.5+7.5(6) =26.5+45$

$\hat{y}=71.5$

The predicted marks for a student studying 6 hours are 71.5 marks.

Q2. Problem: Calculate the Regression Line

X	Y
2	5
4	8
6	12
8	15
10	18

Find:

- $\bar{x}$
- $\bar{y}$
- **b1**
- **b0**
- **Regression equation**
- **Predicted Y when X=12**

Step 1: Calculate Mean of X

$$\overline{x} = \frac{(2+4+6+8+10)}{5}$$

$$\overline{x} = 6$$

Step 2: Calculate Mean of Y

$$\overline{y} = \frac{(5+8+12+15+18)}{5} = 11.6$$

Step 3: Prepare the Calculation Table

X	Y	$X - \overline{x}$	$Y - \overline{y}$	Product	$(X - \overline{x})^2$
2	5	-4	-6.6	26.4	16
4	8	-2	-3.6	7.2	4
6	12	0	0.4	0	0
8	15	2	3.4	6.8	4
10	18	4	6.4	25.6	16
Total				**66**	**40**

Step 4: Calculate Slope b1

- Formula: $b1 = \frac{\sum(X-\bar{x})\,(Y-\bar{y})}{\sum(X-\bar{x})^2}$

Substitute:

- b1=66/40
- b1=1.65

Step 5: Calculate Intercept b0

- Formula: $b0=\bar{y}-b1\bar{x}$

b0=11.6−(1.65)(6)

b0=11.6−9.9

b0= 1.7

Step 6: Regression equation: $\hat{y}$ =b0+b1X

$\hat{y}$ = 1.7 + 1.65X

Step 7: Predict Y when X = 12

$\hat{y}$ =1.7 + (1.65)*12 =1.7+19.8

$\hat{y}$=21.5

Q3:Find the regression equation:

X	Y
1	2
2	4
3	5
4	8
5	10

$\bar{x} = \mathbf{3}$

$\bar{y} = \mathbf{5.8}$

The calculation gives:

$\sum(X-\bar{x})(Y-\bar{y})=20$

$\sum(X-\bar{x})^2=10$

Therefore: b1=20/10=2

Now:

- b0=5.8−(2)(3)
- b0=−0.2

Therefore: $\hat{y}$ = −0.2 + 2X

Predict Y when X = 6

- $\hat{y}$=−0.2+2(6)
- $\hat{y}$=11.8

ORDINARY LEAST SQUARES ESTIMATION

- **Ordinary Least Squares (OLS)** is a method used to estimate the **coefficients of a regression equation**.
- The objective of OLS is to find the regression line that gives the **minimum sum of squared errors** between actual and predicted values.

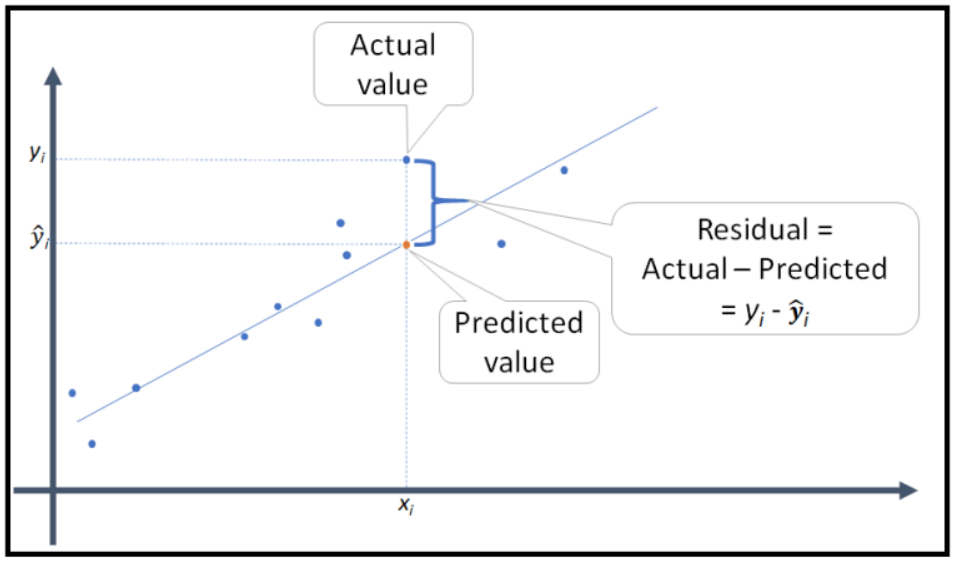
Residual / Error:

- The difference between the actual value and predicted value is called the **residual**.

$$e = Y - \hat{y}$$

Where:

- Y = Actual value
- $\hat{y}$ = Predicted value



Sum of Squared Errors:

$$SSE = \sum (Y - \hat{y})^2$$

OLS tries to minimize SSE.

Q4: The actual and predicted values are.

Calculate SSE.

Actual Y	Predicted $\hat{y}$
10	12
15	14
20	18
25	27
30	29

Solution

- Formula: $SSE=\sum(Y-\hat{y})^2$

Actual	Predicted	Error	Error²
10	12	-2	4
15	14	1	1
20	18	2	4
25	27	-2	4
30	29	1	1

Therefore:

- SSE=4+1+4+4+1
- SSE=14

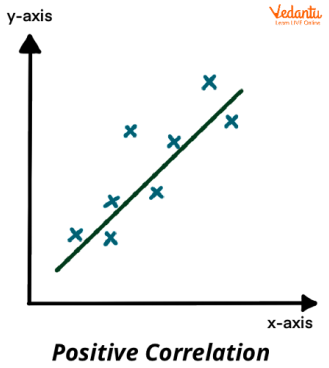
OLS selects the regression coefficients that minimize the sum of squared errors.

CORRELATION

- **Correlation** measures the **strength and direction of the linear relationship between two variables**.
- The Pearson correlation coefficient is represented by: r
- Its value lies between:$-1 \leq r \leq +1$

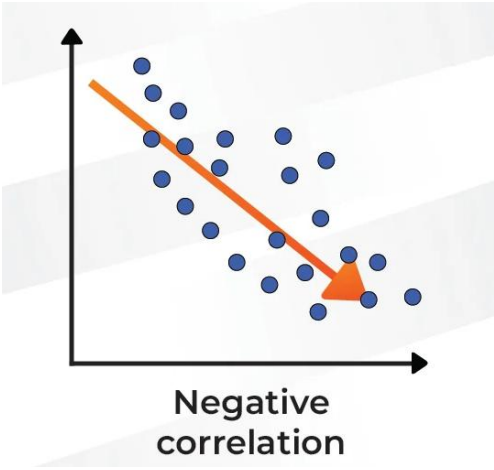
Types of Correlation

1. Positive Correlation:
   - When X increases, Y tends to increase.
   - X ↑ → Y ↑
   - Example:Study hours ↑ → Marks ↑



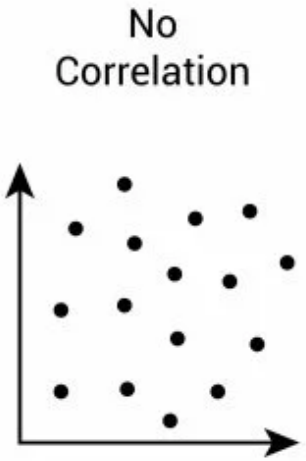
2. Negative Correlation:

- When X increases, Y tends to decrease.
- X ↑ → Y ↓
- Example:Price ↑ → Demand ↓



3. Zero Correlation:

- There is no clear linear relationship between X and Y.



Interpretation of r:

Correlation coefficient	Interpretation
+1	Perfect positive
Close to +1	Strong positive
0	No linear correlation
Close to −1	Strong negative
−1	Perfect negative

Q5. Calculate the correlation coefficient for:

X	Y
1	30
2	35
3	45
4	50
5	60

$\bar{x} = \mathbf{3}$

$\bar{y} = \mathbf{44}$

From the calculation:

$\sum(X-\bar{x})(Y-\bar{y}) = 75$

$\sum (\mathbf{X} - \bar{\boldsymbol{x}})^{\mathbf{2}} = 10$

$\sum (\mathbf{Y} - \bar{\boldsymbol{y}})^{\mathbf{2}} = 570$

Using:

$$r=\frac{\sum(X-\bar{x})(Y-\bar{y})}{\sqrt{\sum(X-\bar{x})^2\sum(Y-\bar{y})^2}}$$

$$r=\frac{75}{\sqrt{10*570}}$$

$$r=\frac{75}{\sqrt{5700}}$$

$$r \approx 0.993$$

Since r is very close to +1

There is a very strong positive linear correlation between X and Y.

MULTIPLE LINEAR REGRESSION

- **Multiple Linear Regression** is used to predict one dependent variable using **two or more independent variables**.
- **General Equation**

$$\hat{y}=b_0+b_1X_1+b_2X_2+\cdots+b_nX_n$$

Where:

- $\hat{y}$= Predicted dependent variable
- $X_1, X_2, \ldots X_n$= Independent variables
- b_0 = Intercept
- $b_1, b_2, \ldots\ldots, b_n$ = Regression coefficients

Example

To predict student marks:

- X_1 = Study Hours
- X_2 = Attendance
- X_3 = Assignment Score

The model may be: $\hat{y}=10 + 4\ X_1 + 0.5\ X_2 + 2\ X_3$

Find the predicted marks for:

- Study Hours = 5
- Attendance = 80
- Assignment Score = 7

Substitute the values:

- Y=10+4(5)+0.5(80)+2(7)
- Y=10+20+40+14
- Y

Predicted marks = 84

$\hat{y}$=10 + 4 X_1 +0.5 X_2 +2 X_3

Intercept = 10

- When all independent variables are zero, the predicted value of Y is 10.

Coefficient of X_1 =4

- For a one-unit increase in study hours, predicted marks increase by **4 units**, keeping the other variables constant.

Coefficient of X_2 =0.5

- For a one-unit increase in attendance, predicted marks increase by **0.5 units**, keeping the other variables constant.

Coefficient of X_3 =2

- For a one-unit increase in assignment score, predicted marks increase by **2 units**, keeping the other variables constant.

SIMPLE VS MULTIPLE LINEAR REGRESSION

Feature	Simple Linear Regression	Multiple Linear Regression
Independent variables	One	Two or more
Equation	$Y=b_0+b_1X$	$Y=b_0+b_1X_1+b_2X_2+\cdots+b_nX_n$
Example	Marks based on study hours	Marks based on study hours, attendance, assignments
Complexity	Lower	Higher
Interpretation	Easier	More complex